# NovaCool Digital Twin

Physics-Based Simulation · Control Policy · RL Environment Design

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Hammerhead — Simulation Engineer Technical Case Study

## System Overview & Design Philosophy

NovaCool is a 10 MW colocation facility: 200 server racks in a hot-aisle/cold-aisle layout served by four Computer Room Air Handlers (CRAHs). The simulation platform must predict thermal conditions and power consumption under varying workloads and serve as a training environment for autonomous control agents.

**Design philosophy.** A CFD model would be accurate but useless here — it cannot run fast enough to serve as an RL training environment. Instead, following Zohdi [1], the twin is built as a lumped-parameter reduced-order model: each rack is a heat source, each CRAH is a heat sink, and the dominant physics (first law of thermodynamics, fan affinity laws, UPS efficiency curves) are captured analytically. A complete 24 hour episode runs in under one second, which is the property that makes autonomous control training viable.

**Codebase architecture.** Three layers, each with a single `step()` interface: a *physics layer* (`ThermalModel`, `PowerModel`), a *control layer* (`FixedSetpointController`, `ThermostatController`), and an *environment layer* (`DataCenterEnv`). The orchestrator calls each layer's `step()` and combines outputs — adding a humidity model or battery storage subsystem requires writing one new class, not touching existing code.

## Part A — Digital Twin (Core Physics)

### A1 — Thermal Model

Each rack $i$ is modelled as a heat-generating device following the first law of thermodynamics. Cold air supplied by the assigned CRAH unit absorbs heat passing through the rack servers:

$$T_{\text{out},i} = T_{\text{in},i} + \frac{\dot{Q}_i}{\dot{m}_i\, c_p} \tag{1}$$

where $\dot{Q}_i$ is rack IT power [W], $\dot{m}_i$ is effective air mass flow rate [kg/s], and $c_p = 1006\,\text{J/(kg K)}$. Cold-aisle inlet temperature accounts for containment imperfections:

$$T_{\text{in},i} = T_{\text{supply,CRAH}(i)} + \underbrace{\delta_{\text{recirc}}}_{1.5°\text{C}} \tag{2}$$

**CRAH cooling model.** Heat removed per unit:

$$\dot{Q}_{\text{CRAH},j} = \dot{m}_{\text{total},j}\, c_p\, (T_{\text{return},j} - T_{\text{supply},j}) \tag{3}$$

where $\dot{m}_{\text{total},j} = \dot{m}_{\text{rack}} \times 50 \times f_{\text{fan},j}$.

**Calibration:** Initial $\dot{m} = 0.36\,\text{kg/s}$ gave $T_{\text{out}} = 132°\text{C}$ — physically impossible. Back-calculated from sensor reference: $\dot{m} = \dot{Q}/(c_p \Delta T) = 45{,}000/(1006 \times 28.5) \approx 1.57\,\text{kg/s}$. Set to **1.5 kg/s** (ASHRAE A2 envelope). Peak outlet corrected: 132°C → 38.3°C.

### A2 — Power Model & PUE

UPS efficiency peaks at 62.5% load (quadratic model):

$$\eta_{\text{UPS}}(x) = \eta_{\text{peak}} - k\,(x - x_{\text{peak}})^2 \tag{4}$$

with $\eta_{\text{peak}} = 0.97$, $x_{\text{peak}} = 0.625$, $k = 0.23$, and $x = P_{\text{IT}}/10{,}000\,\text{kW}$. CRAH fan power follows the affinity cube law:

$$P_{\text{fan},j} = P_{\text{rated}} \cdot f_{\text{fan},j}^3 \qquad P_{\text{rated}} = 50\,\text{kW} \tag{5}$$

$$\text{PUE} = \frac{P_{\text{UPS,in}} + P_{\text{fans}} + P_{\text{pumps}} + P_{\text{overhead}}}{P_{\text{IT}}} \tag{6}$$

**Result:** Baseline mean PUE = **1.134** (range 1.089–1.221). Below the world-class target of 1.2 for over 70% of the 24-hour window.

### A3 — Model Validation & Calibration

Simulation compared against `sensor_reference.csv` across all 200 racks × 1,440 timesteps:

**Table 1:** Outlet & Inlet Temperature Validation Metrics

| Metric | Outlet (°C) | Inlet (°C) |
|---|---|---|
| RMSE | 4.09 | 0.78 |
| MAE | 3.34 | 0.62 |
| MaxAE | 15.48 | 3.36 |

**Divergence analysis.** Error peaks in two windows: **06:12–06:55** and **20:58–21:56** (mean error ≈ 6.5°C at workload shift-changes). Root cause: the steady-state assumption in Eq. (1) holds at quasi-static load but lags during fast transients — analogous to assuming quasi-static loading in structural dynamics.

**Calibration approaches:**

1. **Bayesian parameter tuning:** Treat $\dot{m}_{\text{rack}}$ and $\delta_{\text{recirc}}$ as uncertain priors; update posteriors via MCMC

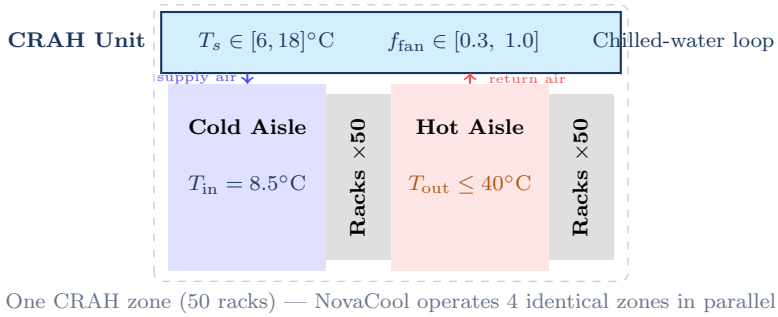


**Figure 1:** Hot-aisle/cold-aisle containment layout for one CRAH zone. Cold supply air enters the cold aisle, passes through server racks absorbing heat, and exits as warm return air to the CRAH unit.

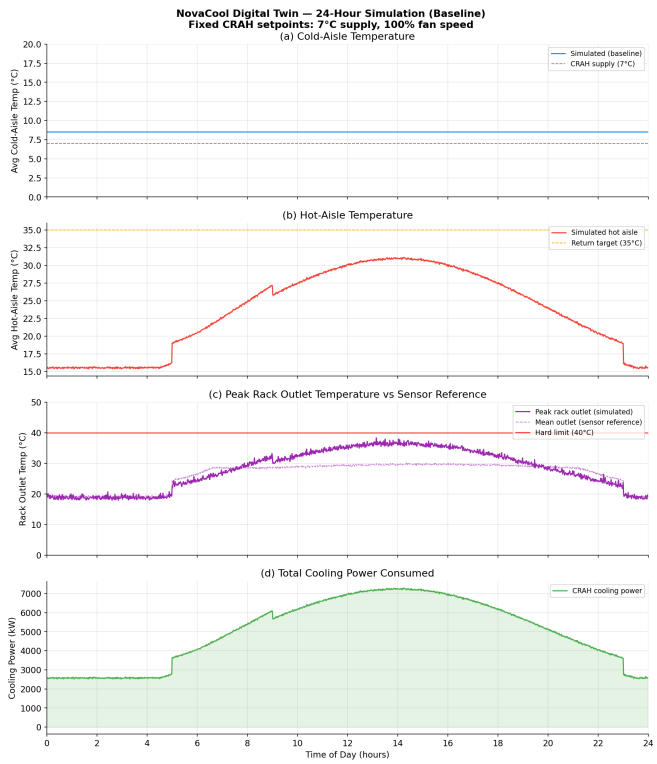


**Figure 2:** 24-hour thermal simulation: cold/hot aisle temperatures, peak rack outlet, and total cooling power (baseline).

against sensor reference. Best for offline commissioning.

2. **Kalman filter assimilation:** At each timestep blend model prediction with live sensor: $\hat{\mathbf{x}}^+ = \hat{\mathbf{x}}^- + K(\mathbf{y} - H\hat{\mathbf{x}}^-)$. Corrects slow parameter drift (e.g. filter fouling) in real time.

## Part B — Control & RL Environment

### B4 — Control Policy

**Baseline.** Fixed setpoints: $T_{\text{supply}} = 7°\text{C}$, $f_{\text{fan}} = 1.0$ for all CRAHs and all timesteps. Open-loop, no feedback — the reference case.

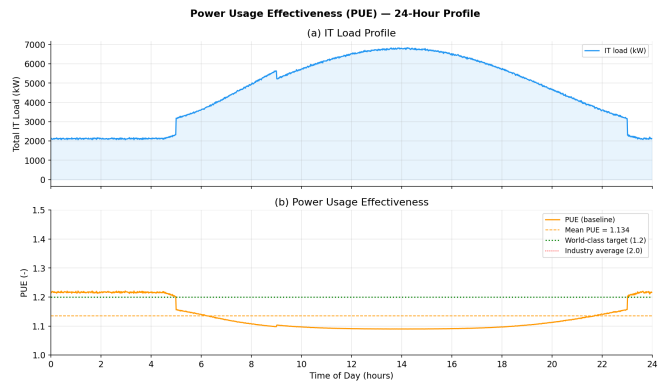


**Figure 3:** PUE profile: below world-class 1.2 target for over 70% of the day at peak load.

**Heuristic controller.** A dead-band thermostat on max rack outlet temperature per CRAH zone, following Zohdi's [2] adaptive control paradigm:

$$T_{s,j}^{t+1} = \begin{cases} T_{s,j}^t - \delta_T & T_{\max,j} > T_{\text{high}}\ (37°\text{C}) \\ T_{s,j}^t + \delta_T & T_{\max,j} < T_{\text{low}}\ (33°\text{C}) \\ T_{s,j}^t & \text{otherwise (dead-band)} \end{cases} \tag{7}$$

with $\delta_T = 0.5°\text{C/step}$, fan speed modulated in parallel.

**Table 2:** Control Policy Comparison — 24-hour Window

| Metric | Baseline | Heuristic |
|---|---|---|
| IT Load Served (MWh) | 108.94 | 108.94 |
| Thermal Violations | 0 | 10 |
| Cooling Energy (MWh) | 119.81 | 117.16 |
| Mean PUE | 1.134 | 1.101 |
| PUE Improvement | — | **3.0%** |

**Honest finding:** The heuristic achieves 3.0% PUE improvement but introduces 10 thermal violations. Root cause: the reactive controller raises supply temperature during low load, then cannot cool fast enough

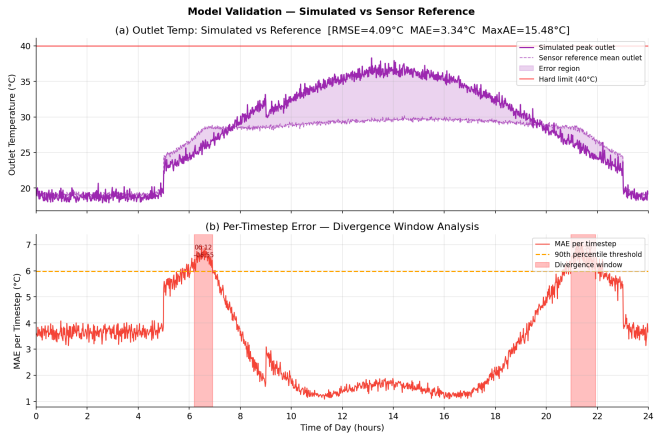


**Figure 4:** Model validation: simulated vs sensor reference with divergence windows at shift-change periods.

during the sharp 05:00 workload ramp. This *directly motivates the RL environment* — a predictive agent observing time-of-day can pre-emptively lower supply temperature before the ramp occurs.

### B5 — RL Environment Design

**Observation space** $s_t \in \mathbb{R}^{13}$, normalised to $[0, 1]$:

| Variable | Description |
|---|---|
| $\bar{T}_{\text{cold}}$, $\bar{T}_{\text{hot}}$, $T_{\max}$ | Zone temperatures |
| $P_{\text{IT}}/P_{\text{cap}}$ | Normalised IT load |
| $t/T$ | Time of day |
| $\mathbf{T}_s \in \mathbb{R}^4$ | CRAH supply temps |
| $\mathbf{f} \in \mathbb{R}^4$ | Fan speed fractions |

**Action space** $a_t \in \mathbb{R}^8$: $\{\Delta T_{s,j}, \Delta f_j\}_{j=1}^4$, clipped to $|\Delta T| \leq 2°\text{C/step}$, $|\Delta f| \leq 0.1/\text{step}$.

**Reward function:**

$$r_t = \underbrace{w_1 P_{\text{IT}}}_{\text{throughput}} - \underbrace{w_2 \max(0, T_{\max} - 40)^2}_{\text{safety}} - \underbrace{w_3 P_{\text{facility}}}_{\text{energy}} \quad (8)$$

$w_1 = 0.005, \quad w_2 = 100, \quad w_3 = 0.001.$

**Reward design justification:** The violation term is *squared* — a linear penalty produces boundary-hugging at exactly 40°C, which is unsafe. The quadratic creates an accelerating gradient keeping the agent well clear. Weight ratio $w_2/w_1 = 20{,}000$ ensures thermal safety dominates: the agent cannot profitably trade violations for throughput. The energy term is a soft secondary objective.

**Episode termination:** Always at $t = 1{,}440$. Early termination on violation was considered and rejected: a truncated episode removes the agent's ability to observe the downstream cost of a violation, which produces biased Q-value estimates and makes training unstable. The squared penalty in Eq. (8) provides a strong enough gradient without truncation.

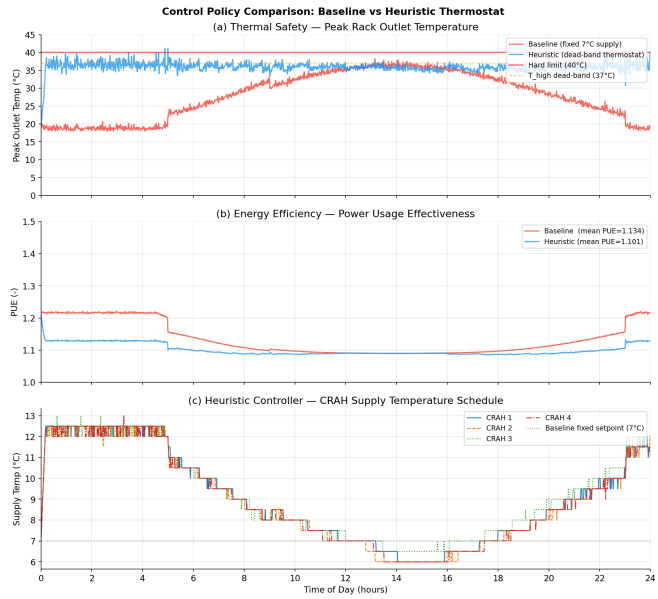


**Figure 5:** Control comparison: heuristic achieves 3.0% PUE improvement; 10 violations motivate RL solution.

**Validation:** Null-action rollout (zero delta, 7°C fixed) gives total reward = 25,384, final PUE = 1.216, max outlet = 19.18°C — confirms the reward signal is positive under safe conservative operation.

## Part C — Architecture & Scaling

**Modularity.** Every subsystem implements a single `step(input) -> output` contract. The `NovaCoolFacility` orchestrator knows nothing about the internals of `ThermalModel` or `PowerModel` — it only calls `step()` and combines the returned dicts. Adding humidity modelling, battery storage, or a new cooling technology means writing one new class that satisfies the contract. Nothing existing breaks.

**Scaling to 2,000 racks.** The current model is fully vectorised with NumPy — the per-rack thermal loop is a single array operation, so 10× more racks adds negligible wall-clock time. For larger deployments: (1) *zonal aggregation* reduces state dimensionality by grouping racks into thermal clusters; (2) *surrogate models* (small neural nets trained on simulation outputs) enable $\mathcal{O}(\mu s)$ rollout during RL training, following Zohdi [3].

**Online correction (sim-to-real alignment).** At each minute, a Kalman filter blends the model prediction with incoming sensor readings: $\hat{\mathbf{x}}^+ = \hat{\mathbf{x}}^- + K(\mathbf{y} - H\hat{\mathbf{x}}^-)$. Gain $K$ is tuned offline against historical data, correcting slow parameter drift such as filter fouling changing airflow resistance.

**Testing strategy.** Unit tests (14/14 passing) verify: first-law energy conservation, UPS cube-law fan power, Gym API contract (shape, episode length, normalisation,

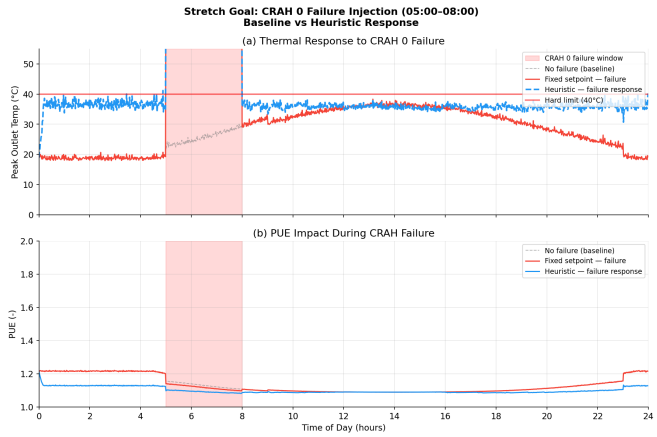


**Figure 6:** Stretch goal: CRAH 0 failure injection (05:00–08:00). Heuristic controller partially compensates; RL agent could learn proactive load redistribution.

reward sign). Integration tests pin the full 24-hour baseline run as a golden file — any regression is immediately caught. Property-based testing (Hypothesis) can fuzz-test the thermal model over the full physically plausible input space.

**Sim-to-real gap — top sources and mitigations:**

1. **Hot-aisle bypass recirculation:** The recirculation factor $\delta_{\text{recirc}}$ is a global scalar; real bypass varies per rack location. *Mitigation:* per-zone coefficients from in-situ smoke visualisation tests.
2. **CRAH thermal lag:** CRAH response is modelled as instantaneous; real units have ∼5-minute thermal mass. *Mitigation:* add a first-order lag ODE state to the CRAH model, calibrated via step-response test.
3. **Rack airflow non-uniformity:** Servers within a rack pull unequal airflow by chassis density. *Mitigation:* per-rack $\dot{m}$ multipliers calibrated from PDU power and inlet sensor correlation.

**Domain randomisation & partial observability.** A heuristic trained on a single parameter set will fail when deployed on a real facility where $\dot{m}_{\text{rack}}$ and $\delta_{\text{recirc}}$ differ from the simulation defaults. The fix is to randomise these per episode during training: $\dot{m}_{\text{rack}} \sim \mathcal{U}(1.2, 1.8)\,\text{kg/s}$, $\delta_{\text{recirc}} \sim \mathcal{U}(1.0, 2.5)°\text{C}$, workload burst magnitude varied ±20%. The agent never sees the same facility twice and is forced to learn a policy that works across the range. Sensor failures are modelled by randomly masking observation dimensions per episode — the agent learns to act under partial observability without any interface changes.

## References

[1] T. I. Zohdi, "A digital-twin and machine-learning framework for precise heat and energy management of data-centers," *Computational Mechanics*, 2022. DOI: 10.1007/s00466-022-02152-3

[2] T. I. Zohdi, "An adaptive digital framework for energy management of complex multi-device systems," *Computational Mechanics*, 2022. DOI: 10.1007/s00466-022-02212-8

[3] T. I. Zohdi, "A machine-learning digital-twin for rapid large-scale solar-thermal energy system design," *Comput. Methods Appl. Mech. Eng.*, 412, 2023.